

MASON COUNTY AP, WV, USA

WMO: 742079

Lat: 38.915N

Lon: 82.099W

Elev: 643

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 06-18

WBAN: 63876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	13.5	-1.6	5.2	14.6	3.1	6.6	15.9	17.5	41.5	15.7	41.9	1.7	280	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	91.3	73.1	90.3	72.7	88.2	72.0	77.4	87.6	76.1	85.6	74.6	83.5	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	131.6	82.1	72.9	125.2	80.4	72.2	122.0	79.3	41.1	86.9	39.9	84.8	38.5	84.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.0	12.1	10.5	1.7	95.3	4.8	2.4	-1.8	97.0	-4.6	98.4	-7.3	99.7	-10.8	101.5		
			2.1	79.3	5.5	2.7	-1.9	81.3	-5.1	82.9	-8.2	84.4	-12.2	86.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	32.9	35.3	45.1	55.9	65.8	72.4	74.3	73.9	67.5	55.7	44.4	37.6
	DBStd	17.18	11.53	11.38	9.98	9.01	7.83	5.25	4.74	4.43	6.87	8.76	8.14	9.47
	HDD50	1878	543	429	216	43	2	0	0	0	0	40	206	399
	HDD65	4666	996	832	619	291	91	8	1	1	49	310	618	850
	CDD50	3767	12	17	65	221	491	673	753	740	524	218	38	15
	CDD65	1080	0	1	3	19	116	231	288	276	123	23	0	0
	CDH74	10759	0	7	47	384	1314	2299	2632	2509	1255	307	4	1
	CDH80	4084	0	0	9	97	466	900	1070	996	469	77	0	0
Wind	WSAvg	3.0	4.0	4.0	3.8	4.2	2.8	2.5	2.0	1.6	1.8	2.5	2.9	3.4
Precipitation	PrecAvg	43.0	3.4	2.9	4.0	3.4	4.7	4.2	4.2	3.6	3.6	2.6	2.9	3.5
	PrecMax	53.6	6.3	5.6	9.0	6.2	8.1	10.0	5.4	5.6	9.6	7.3	6.0	8.5
	PrecMin	32.9	1.5	0.8	1.4	1.3	0.6	1.4	2.6	1.3	0.7	0.7	1.4	1.9
	PrecStd	5.8	1.3	1.3	1.7	1.4	2.4	1.9	0.7	1.2	2.4	1.6	1.1	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	72.9	80.5	87.9	90.7	93.4	93.5	95.4	93.2	86.5	73.4	70.0
		MCWB	58.2	59.3	62.5	63.2	71.9	73.0	76.1	73.6	70.3	68.0	59.2	57.9
	2%	DB	61.7	64.4	72.7	82.2	88.2	90.6	91.0	90.9	90.0	81.6	69.7	63.8
		MCWB	55.4	54.7	57.5	61.8	70.5	73.5	74.5	72.4	70.5	65.0	56.0	55.1
	5%	DB	56.8	59.3	67.8	78.9	85.9	88.3	89.7	89.6	86.0	77.1	64.2	58.7
		MCWB	52.5	52.7	54.4	60.6	69.1	72.5	74.2	72.4	70.6	63.0	54.0	50.5
	10%	DB	51.7	54.3	62.6	73.2	81.8	86.0	86.3	86.3	81.5	72.1	60.8	53.9
		MCWB	46.9	48.3	51.6	58.0	67.0	71.1	72.5	71.2	67.1	61.6	52.0	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	62.9	64.3	68.5	75.5	78.3	80.2	78.3	76.4	72.0	62.3	61.2
		MCDB	63.9	70.5	76.4	78.5	86.0	88.5	90.2	88.2	88.7	77.7	70.2	67.2
	2%	WB	56.6	58.0	60.9	65.3	73.3	76.4	77.9	76.7	74.0	68.8	58.8	56.7
		MCDB	61.2	63.4	68.7	76.7	84.1	86.4	87.8	86.4	84.1	77.9	65.2	62.4
	5%	WB	52.9	54.0	57.9	62.9	71.4	74.8	76.3	75.1	72.1	65.5	55.6	52.7
		MCDB	57.0	59.0	63.6	72.4	81.5	84.2	85.5	84.0	81.1	74.8	62.7	56.5
	10%	WB	46.4	48.0	54.1	60.7	69.2	73.2	74.6	73.5	70.1	62.3	53.2	47.8
		MCDB	50.4	53.5	60.7	70.5	78.1	82.0	82.7	81.8	77.0	69.7	59.2	52.9
Mean Daily Temperature Range	5% DB	MDBR	16.8	18.6	21.7	25.8	24.6	23.6	22.5	23.8	24.7	23.7	22.1	16.5
		MCDBR	22.8	27.5	30.5	33.7	29.2	27.2	25.6	27.4	28.4	30.6	29.3	25.5
		MCWBR	17.3	19.6	16.7	15.8	13.0	11.7	10.5	11.4	11.9	15.5	18.3	18.3
	5% WB	MCDBR	20.8	25.6	23.3	26.1	25.1	22.2	21.8	21.9	24.4	26.7	24.3	22.1
		MCWBR	16.8	19.7	15.1	14.5	11.8	10.8	10.1	10.1	11.3	13.8	17.1	18.8
Clear-Sky Solar Irradiance	taub	0.311	0.325	0.352	0.396	0.441	0.464	0.486	0.485	0.407	0.365	0.330	0.302	
	taud	2.480	2.419	2.405	2.294	2.182	2.164	2.120	2.101	2.301	2.422	2.493	2.529	
	Ebn at Noon	275	285	287	279	268	261	253	250	265	266	265	270	
	Edh at Noon	25	30	34	41	46	47	49	49	38	30	25	23	
All-Sky Solar Radiation	RadAvg	556	784	1136	1521	1734	1912	1813	1706	1414	1004	702	482	
	RadStd	56	113	95	130	148	114	99	100	137	111	90	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (27 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+117

Nomenclature: See separate page